BHUSHAN WALUNJ

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PROFESSIONAL SUMMARY

AI & RPA Engineer pursuing B.E. in Electronics & Telecommunication with hands-on experience in Python, Generative AI, RAG pipelines, Computer Vision, Workflow Automation, and Robotic Process Automation. Skilled in LangChain, YOLOv5, n8n, and AutomationEdge. Seeking AI/ML, RPA, or Intelligent Automation roles.

TECHNICAL SKILLS

AI / ML: PyTorch · YOLOv5 · YOLOv8 · OpenCV · LangChain · Hugging Face · LLMs · RAG · Computer Vision · Deep Learning · Generative AI · Model Deployment

RPA & Auto: AutomationEdge · n8n · Intelligent Automation · Process Automation · AI Agents · Workflow Orchestration · REST APIs · JSON · Webhooks

Programming: Python (Intermediate) · C++ · MATLAB · Basic Java

Deployment: Flask · FastAPI · Git · Streamlit · VS Code

Data & CS: NumPy · Pandas · Matplotlib · DSA · DBMS · Digital Circuits

Hardware: Verilog HDL · VLSI Fundamentals · Microcontrollers (ESP32, Arduino UNO)

PROJECTS

CityFix AI — Multi-Agent Civic Issue Reporting System 2026 – Present

LangGraph · Groq (Llama 3.3 70B) · Streamlit · SQLite · Plotly

- Built an AI app where users report civic problems (like potholes or broken streetlights) with a photo, and multiple AI agents work together to identify the issue, sort it by category, and route it automatically.
- Added location detection to pin each issue on a map and auto-generate a PDF report — cutting manual reporting work.

LoanDesk — RAG-Based Loan Advisory Chatbot 2025 – Present

LangChain · ChromaDB / FAISS · Python · Streamlit · LLM API

- Built a RAG pipeline for context-aware Q&A over loan policy documents, achieving sub-2s response latency with LangChain + ChromaDB vector retrieval.
- Designed Streamlit front-end for PDF upload and conversational querying — end-to-end AI system design.

AI-Based Helmet Violation Detection System (B-DESK) 2024 – 2025

Python · YOLOv5 · PyTorch · Flask · OpenCV

- Trained custom YOLOv5 model with optimised anchors and data augmentation; deployed as Flask app supporting static image and live webcam prediction.
- Integrated real-time OpenCV inference pipeline with optimised frame pre-processing for smooth live detection.

AI Agent & Workflow Automation Platform 2025 – Present

n8n · Python · REST APIs · LLM Integration · Webhooks

- Designed multi-step AI workflows in n8n integrating LLM nodes with REST APIs for intelligent document routing and enterprise automation.

Verilog HDL — Digital Logic Design 2024

Verilog HDL · ModelSim

- Implemented Full Adder, 4:1 MUX, and FSM circuits in Verilog HDL with testbench-driven simulation and waveform verification.

EXPERIENCE

RPA Trainee Intern · ValueDX Technologies, Pune June 2026 – Present

- Building enterprise RPA workflows using AutomationEdge Process Studio — JSON, REST APIs, document processing, and bot deployment.
- Applying exception handling, debugging, and automation best practices for production-grade bots integrating AI and RPA.

Generative AI & Agentic Systems Intern · Lenovo LEAP × AICTE (Remote) June 2026 – July 2026

- Developing LLM-based applications, prompt engineering, and agentic AI pipelines as part of Lenovo LEAP × AICTE national virtual internship.

Python Developer Intern · MASS IT Solutions LLP, Pune Jan 2026 – Feb 2026

- Built AI-driven CV modules using Python, PyTorch, YOLOv5; optimised end-to-end model training pipelines (preprocessing → augmentation → evaluation → inference).
- Applied quantitative validation and performance tuning, reducing inference latency in real-time production-adjacent environments.

EDUCATION

B.E. — Electronics & Telecommunication | Current SGPA: 8.52 · SKNCOE, Pune Aug 2023 – Present

HSC (Science) | Grade: B · GRPS College of Science, Pune · Maharashtra State Board 2021 – 2023

CERTIFICATIONS

AutomationEdge Foundation Certification · AutomationEdge Academy	2026
AutomationEdge Intermediate Certification · AutomationEdge Academy	2026
Gen AI Engineering Mastermind · Outskill (AI Researcher @ Silival, Data Scientist @ SLK)	2026
Gen AI 101 · FutureSkills Prime · IT-ITeS SSC Nasscom (MeitY — Govt. of India recognised)	Apr 2026
MathWorks MATLAB Onramp · MathWorks	Oct 2025
Deloitte Australia — Cyber Security Job Simulation · Forage	Jul 2025
Deloitte Australia — Data Analytics Job Simulation · Forage	Jul 2025